



1. [10 pts.] Write out the form of the partial fraction decomposition of the function (Do not determine the numerical values of the coefficients): $\frac{x^4 + 2x^2 + 7}{(x^3 - x^2 + x)^2(x^2 - 1)}$.

Solution: Since $(x^3 - x^2 + x)^2(x^2 - 1) = (x(x^2 - x + 1))^2(x - 1)(x + 1) = x^2(x^2 - x + 1)^2(x - 1)(x + 1)$, we deduce that

$$\frac{x^4 + 2x^2 + 7}{(x(x^2 - x + 1))^2(x - 1)(x + 1)} = \frac{A_1}{x} + \frac{A_2}{x^2} + \frac{A_3x + A_4}{x^2 - x + 1} + \frac{A_5x + A_6}{(x^2 - x + 1)^2} + \frac{A_7}{x - 1} + \frac{A_8}{x + 1}.$$

2. [15 pts.] Find $\lim_{x \rightarrow \infty} \left(\frac{3x - 2}{3x + 1} \right)^{2x - 1}$.

The limit is indeterminate of the form 1^∞ . Let $y = \left(\frac{3x - 2}{3x + 1} \right)^{2x - 1}$. This implies that

$$\ln y = (2x - 1) \ln \left(\frac{3x - 2}{3x + 1} \right) = \frac{\ln \left(\frac{3x - 2}{3x + 1} \right)}{\frac{1}{2x - 1}}. \text{ Then, we obtain: } \lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{\ln \left(\frac{3x - 2}{3x + 1} \right)}{\frac{1}{2x - 1}}$$

which is indeterminate of the form $0/0$. Invoking l'Hospital's rule, we have:

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \left(\frac{\frac{3x + 1}{3x - 2} \cdot \frac{(3x + 1)3 - (3x - 2)3}{(3x + 1)^2}}{-2/(2x - 1)^2} \right) = -\frac{9}{2} \lim_{x \rightarrow \infty} \left(\frac{(2x - 1)^2}{(3x - 2)(3x + 1)} \right) = -2.$$

Therefore $\lim_{x \rightarrow \infty} y = e^{-2}$.

3. [15 pts.] Evaluate $\int (\ln(x))^2 dx$.

Solution: We integrate by parts with $u = (\ln(x))^2$ and $dv = dx$, which means that $du = \frac{2 \ln x}{x} dx$ and $v = x$. Then, we obtain:

$$\int (\ln(x))^2 dx = x(\ln(x))^2 - \int x(2 \ln x) \frac{1}{x} dx = x(\ln(x))^2 - 2 \int \ln x dx.$$

Another integration by parts implies that $\int \ln x dx = x \ln x - x + C$ and hence

$$\int (\ln(x))^2 dx = x(\ln(x))^2 - 2(x \ln x - x) + C.$$

4. [15 pts.] Find $\int \frac{\sec^3(x) \sqrt{\tan(x)}}{\cos(x)} dx$.

Solution: Using the fact $1/\cos x = \sec x$, the integral can be written as follows:

$$\int \frac{\sec^3(x) \sqrt{\tan(x)}}{\cos(x)} dx = \int \sec^4 x \tan^{1/2} x dx = \int \sec^2 x \tan^{1/2} x \sec^2 x dx.$$

Next, using the identity $\sec^2 x = 1 + \tan^2 x$ and letting $u = \tan x$, the integral becomes:

$$\int (1 + \tan^2 x) \tan^{1/2} x \sec^2 x dx = \int (1 + u^2) u^{1/2} du = \int (u^{1/2} + u^{5/2}) du = \frac{2}{3} u^{3/2} + \frac{2}{7} u^{7/2} + C$$

$$= \frac{2}{3} \tan^{3/2} x + \frac{2}{7} \tan^{7/2} x + C.$$

5. [15 pts.] Find $\int \frac{3x-1}{\sqrt{x^2-6x+13}} dx$.

Solution: We have: $x^2 - 6x + 13 = (x-3)^2 + 4$. Now, let $x-3 = 2 \tan \theta$, $\theta \in (-\pi/2, \pi/2)$.

The integral can be written as follows: $\int \frac{3x-1}{\sqrt{x^2-6x+13}} dx = \int \frac{2(4+3 \tan \theta)}{2 \sec \theta} 2 \sec^2 \theta d\theta$

$$= 2 \int (4 \sec \theta + 3 \sec \theta \tan \theta) d\theta = 2(4 \ln |\sec \theta + \tan \theta| + 3 \sec \theta) + C. \text{ Finally, we have:}$$

$$\int \frac{3x-1}{\sqrt{x^2-6x+13}} dx = 8 \ln |\sqrt{x^2-6x+13} + x - 3| + 3\sqrt{x^2-6x+13} + K,$$

where $K = C - 8 \ln 2$.

6. [15 pts.] Evaluate $\int \frac{(2^{3x} - 4(2^x) - 10)2^x}{2^{2x} - 2^x - 2} dx$.

Solution:

Let $t = 2^x$ and so $dt = 2^x \ln 2 dx$. The integral becomes:

$$\frac{1}{\ln 2} \int \frac{t^3 - 4t - 10}{t^2 - t - 2} dt = \frac{1}{\ln 2} \int \left(t + 1 - \frac{t+8}{(t-2)(t+1)} \right) dt \text{ after performing a long division.}$$

Then, partial fractions gives: $\frac{t+8}{(t-2)(t+1)} = \frac{10/3}{t-2} + \frac{-7/3}{t+1}$. Therefore, $\int \frac{t^3 - 4t - 10}{t^2 - t - 2} dt =$

$$\frac{t^2}{2} + t - \frac{10}{3} \ln |t-2| + \frac{7}{3} \ln |t+1| + C. \text{ Finally, we get:}$$

$$\int \frac{(2^{3x} - 4(2^x) - 10)2^x}{2^{2x} - 2^x - 2} dx = \frac{1}{\ln 2} \left(\frac{2^{2x}}{2} + 2^x - \frac{10}{3} \ln |2^x - 2| + \frac{7}{3} \ln (2^x + 1) \right) + C.$$

7. [15 pts.] Determine whether the integral $\int_0^\infty \frac{\sqrt{x+1}}{2x+3} dx$ is convergent or divergent.

Solution: The integral is improper since the interval is $[0, \infty)$. We have:

$$\int_0^\infty \frac{\sqrt{x+1}}{2x+3} dx = \lim_{b \rightarrow \infty} \int_0^b \frac{\sqrt{x+1}}{2x+3} dx. \text{ In order to evaluate the integral, one can take}$$

$$u = \sqrt{x+1} \text{ and hence } u^2 = x+1, 2u du = dx. \text{ Thus } \int \frac{\sqrt{x+1}}{2x+3} dx = \int \frac{u}{2u^2+1} 2u du$$

$$= \int \left(1 - \frac{1}{2u^2+1} \right) du = u - \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2}u) + C = \sqrt{x+1} - \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2}\sqrt{x+1}) + C.$$

Thereby

$$\int_0^\infty \frac{\sqrt{x+1}}{5+4x} dx = \lim_{b \rightarrow \infty} \left(\sqrt{b+1} - \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2}\sqrt{b+1}) \right) - 1 + \frac{1}{\sqrt{2}} \tan^{-1} \sqrt{2} = \infty$$

and the improper integral is divergent.